

## Computer Lab 1 - GP Regression and Classification

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The labs are the only examination, so you should do the labs **individually**.

You can use any programming language you prefer, but do **submit the code**.

Submit a readable report in:

- **PDF** (no Word documents!)
  - **JuPyteR/Quarto notebook compiled to PDF/HTML**
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### 1. *Homoscedastic GP regression*

- (a) Consider the following GP regression

$$y_i = f(\mathbf{x}_i) + \varepsilon_i, \quad \varepsilon_i \sim N(0, \sigma_\varepsilon^2) \\ f(\mathbf{x}) \sim \text{GP}(0, k(\mathbf{x}, \mathbf{x}'))$$

with a squared exponential kernel

$$k(x, x') = \sigma_f^2 \cdot \exp\left(-\frac{(x - x')^2}{2\ell^2}\right).$$

Write your own code to compute the posterior distribution of  $\mathbf{f}_*$

$$p(\mathbf{f}_* | \mathbf{y}, \mathbf{X}, \mathbf{X}_*)$$

where  $\mathbf{y}$  ( $n \times 1$ ) is the training response data and  $\mathbf{X}$  ( $n \times p$ ) is the training covariate data for  $p$  covariates,  $\mathbf{X}_*$  ( $n_* \times p$ ) is a matrix of test inputs and  $\mathbf{f}_*$  ( $n_* \times 1$ ) is the corresponding function values at the test points. Let the kernel hyperparameters  $\sigma_f$  and  $\ell$ , and the noise standard deviation  $\sigma_\varepsilon$  be inputs to your posterior function.

- (b) Use your code to analyze the **Lidar** data (available on the course web page) with the **Distance** variable as the only covariate/feature in both the mean and variance. Standardize **Distance** to take values in the unit interval  $[0, 1]$ . Set  $\ell = 1$ ,  $\sigma_f = 0.5$  and  $\sigma_\varepsilon = 0.05$  (but also play around to learn!). In particular, do a scatterplot of the data and overlay:
- the posterior mean of  $f(\cdot)$  (computed over a suitable grid)
  - 95% credible bands for  $f(\cdot)$
  - 95% predictive bands for  $y$ .

## 2. Heteroscedastic GP regression

- (a) Consider the following heteroscedastic GP regression

$$\begin{aligned}y_i &= f(\mathbf{x}_i) + \varepsilon_i, \quad \varepsilon_i \sim N(0, \sigma_{\varepsilon,i}^2) \\f(\mathbf{x}) &\sim \text{GP}(0, k(\mathbf{x}, \mathbf{x}')) \\ \log \sigma_{\varepsilon,i}^2 &= w_0 + \mathbf{w}_1^T \mathbf{x}_i\end{aligned}$$

Implement an algorithm that samples from the joint posterior

$$p(\mathbf{f}_*, w_0, \mathbf{w}_1 | \mathbf{y}, \mathbf{X}, \mathbf{X}_*) = p(\mathbf{f}_* | w_0, \mathbf{w}_1, \mathbf{y}, \mathbf{X}, \mathbf{X}_*) p(w_0, \mathbf{w}_1 | \mathbf{y}, \mathbf{X}, \mathbf{X}_*).$$

Use the prior  $(w_0, \mathbf{w}_1)^T \sim N(0, \tau^2 I)$ , independent of  $f$ .

**Hint:**  $p(\mathbf{f}_* | w_0, \mathbf{w}_1, \mathbf{y}, \mathbf{X}, \mathbf{X}_*)$  is really close to the formulas (2.22) to (2.24) in the GPML book, and to what you coded in Problem 1 above. The distribution

$$p(w_0, \mathbf{w}_1 | \mathbf{y}, \mathbf{X}, \mathbf{X}_*)$$

is available in closed form and its expression is close to an expression in the GPML book (if you think a little ...). However,  $p(w_0, \mathbf{w}_1 | \mathbf{y}, \mathbf{X}, \mathbf{X}_*)$  is not a standard (known) distribution, and you need to use MCMC or HMC, or something else, to simulate from it. It is OK to use a package for MCMC/HMC.

- (b) Re-analyze the Lidar data using the heteroscedastic GP regression. Plot the marginal posterior distributions for  $w_0$  and  $w_1$ . Do a similar plot as the one in 1(b) using the heteroscedastic model.

HAVE FUN!